Dual-Rate Routed Attention for Efficient Long-Horizon Time-Series Forecasting

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Abstract

Long-horizon time-series forecasting with Transformers is dominated by the cost of quadratic attention over long lookback windows. Compressed-and-indexed attention—mechanisms that compress the key/value (KV) stream and route each query to a small set of compressed blocks—promises to reduce this cost, but its design space is rarely studied systematically. We treat this family as a two-axis design problem: the KV compression rate c_{KV} and the routing resolution c_R. On the ETTh1 dataset under a frozen direct-96-step forecasting protocol, we evaluate all four combinations of $c_{KV}, c_R \in \{4{:}1, 8{:}1\}$ in a controlled, parameter-equalized 2×2 factorial design with five paired seeds. We report three findings. First, *every* compressed configuration outperforms full causal attention on validation MSE (-18.5% to -29.1%, and in 20/20 paired seed comparisons) at an equal or slightly smaller parameter count; a bit-exact wiring audit confirms the full-attention baseline is correctly implemented. Second, coarsening the router (c_R: 4:1 $\rightarrow$ 8:1) systematically deteriorates accuracy ($+0.046$ validation MSE), while coarsening KV compression (c_{KV}: 4:1 $\rightarrow$ 8:1) is mildly beneficial (-0.043): directional evidence that routing granularity, not KV content density, carries accuracy. Third, the best cell (c_{KV}=8, c_R=4) reduces the attention matrix to 48 columns versus 256 for full attention (5.3$\times$) at a matching parameter count. All statistics are descriptive (five seeds; no significance tests); multi-dataset, significance, and latency evaluations are ongoing work.

1 Introduction

Long-horizon time-series forecasting—predicting the next H steps from a lookback window of length L—is a core task in energy, traffic, and finance applications. Transformer-based forecasting models (Zhou et al., 2021; Nie et al., 2023; Liu et al., 2024) are accurate but pay a quadratic cost $O(L^2)$ in the attention mechanism as the lookback grows. One widely used remedy is *compressed-and-indexed* attention: compress the L key/value (KV) positions into $C = L/c_{KV}$ blocks, and have each query attend to only a small, *routed* subset of those blocks plus a local sliding window. This family underlies recent efficient-attention designs for large language models (Yuan et al., 2025; DeepSeek-AI, 2026), and its components—content-based routing (Roy et al., 2021), low-rank KV compression (DeepSeek-AI et al., 2024)—are individually well known.

What is not well studied is the *design space* of this family: how the two independent compression axes interact. We identify two axes:

- **KV compression rate** c_{KV}: how aggressively the content stream is compressed into blocks ($C = L/c_{KV}$ blocks);
- **Routing resolution** c_R: how finely the router represents the history when deciding which blocks to attend to ($M = L/c_R$ index tokens).

Coarsening either axis cuts compute, but the two axes may not trade accuracy equally. Our central question is therefore a controlled, empirical one: *which axis carries accuracy, and which axis only buys efficiency?*

This paper is a **design study**, not a proposal of a new attention mechanism: every building block we use is a configuration of known ideas (Section 3), and overlaps with concurrent work (Yuan et al., 2025; DeepSeek-AI, 2026) are acknowledged rather than elided (Section 2). Our contributions are:

(a) **A controlled 2×2 factorial design.** All four combinations of $c_{KV}, c_R \in \{4{:}1, 8{:}1\}$ are evaluated on a frozen task—ETTh1, direct 96-step forecasting—under an *auxiliary-identical* contract (identical window, top-k, heads, hidden size, optimizer, schedule) with attention-parameter equalization to within 0.3%, across five paired seeds (Section 4).

(b) **A decoupled-rate finding (directional).** Coarsening the router deteriorates accuracy ($+0.046$ validation MSE for c_R:4:1 $\rightarrow$ 8:1), while coarsening KV compression is mildly beneficial (-0.043). The best cell (c_{KV}=8, c_R=4) reduces the attention matrix 5.3$\times$ relative to full attention at equal parameters. We report these as directional observations: effect sizes are smaller than the within-cell seed standard deviation, and no significance test has been run.

(c) **An honest efficiency/accuracy characterization.** A surprising—and wiring-verified—baseline result: all compressed configurations beat full causal attention on the frozen task (-18.5% to -29.1% validation MSE), with explicit limitations (single dataset, CPU-scale models, descriptive statistics).

(d) **Reproducibility artifacts.** A frozen task specification, machine-readable results manifests, a verified parameter registry, causality-verification suites, and a codebase in which every reported number traces to a file (Appendix C).

Scope and non-claims. This work does not claim a novel attention mechanism, does not claim state-of-the-art forecasting accuracy (no external baselines are run yet), does not claim statistical equivalence or significance (descriptive statistics only, five seeds), and does not report wall-clock speedups: the pre-freeze latency figures from earlier stages of this project are superseded and are deliberately *not* used here. Measured latency, FLOPs, and hardware transfer are planned follow-up work (Section 5).

2 Related Work

2.1 Routed and compressed attention

The general idea of content-based routing goes back to the Routing Transformer (Roy et al., 2021), which clusters tokens with online k-means and restricts attention to tokens in the same cluster. Native Sparse Attention (NSA) (Yuan et al., 2025) combines coarse-grained token compression with fine-grained token selection, making it the closest conceptual cousin of the family studied here. DeepSeek-V4's hybrid attention (DeepSeek-AI, 2026) pairs a Compressed Sparse Attention (CSA) stage with a Heavily Compressed Attention (HCA) stage for million-token contexts, reporting 27% of single-token inference FLOPs and 10% of KV cache versus DeepSeek-V3.2; we adopt the working name "Dual-Rate Routed Attention" (DRRA) for our family in part to avoid colliding with that label. The low-rank KV bottleneck and shared-query projection used in our configurations descend from Multi-head Latent Attention (MLA) (DeepSeek-AI et al., 2024). Attention distillation toward cheaper inference architectures (Goldstein et al., 2025) overlaps a secondary, separable thread of this project (Appendix B) rather than the routing/compression core studied here.

Crucially for our claim (b), none of these works, to our knowledge, *systematically varies the content-compression rate and the routing resolution independently and reports their separate main effects* on a fixed task. The closest are Simplified Sparse Attention (Mao et al., 2026), which compresses context into gist tokens and selectively unfolds top-k chunks but derives k from the chunk size to hold the total budget constant (Eq. A.1), coupling the two axes; and The Sparse Frontier (Nawrot et al., 2025), which sweeps selection budgets across six methods with no content-compression knob. DeepSeek-V4 (DeepSeek-AI, 2026) and NSA (Yuan et al., 2025) expose both mechanisms as fixed hyperparameters without varying them independently. We confirmed this on a systematic sweep of the 2024–2026 routing/compression-attention literature (August 2026).

2.2 Efficient and accurate time-series forecasting

Long-horizon forecasting architectures include the ProbSparse-attention Informer (Zhou et al., 2021), patching-based PatchTST (Nie et al., 2023), the inverted iTransformer (Liu et al., 2024), linear baselines (Zeng et al., 2023), and 2D-variation modeling with TimesNet (Wu et al., 2023). These are the natural external baselines for this line of work; none has been run under our frozen protocol yet **[TODO: Gate 3: run PatchTST / iTransformer / DLinear / Informer (and a linear and an SSM baseline) under the frozen protocol; collect their citations with the runs (plan/next_step.md §7)]**. Our contribution is orthogonal: we do not propose a new forecasting architecture but study one design axis of the attention module that any of these models could use.

2.3 Positioning summary

This paper sits in the intersection of efficient attention and time-series forecasting, as a *controlled design study*: it measures how two configurable axes of a fixed attention family trade off accuracy and efficiency on one frozen task, and it does so with the reproducibility discipline (pre-registered factorial, paired seeds, parameter equalization, frozen task spec) that makes the trade-off structure measurable rather than anecdotal.

3 Method

3.1 Task and evaluation protocol (frozen)

All experiments use a single frozen task specification **[TODO: cite plan/gate1_task.md]**. The dataset is ETTh1 (Electricity Transformer Temperature, hourly; 7 features), split chronologically $60\%/20\%/20\%$ into train/validation/test with stride-1 sliding windows of length $L{=}256$ that never span a split boundary. Normalization is a per-feature z-score whose statistics are fit on the *train* split only and applied to all splits (the earlier practice of fitting over the full series was a future-statistics leak and is fixed). The task is *direct* 96-step forecasting: in one forward pass the model maps $x \in \mathbb{R}^{B\times 256\times 7}$ to $\hat{y} \in \mathbb{R}^{B\times 96\times 7}$, and the loss is the pooled MSE over all $(B, 96, 7)$ elements, reported in standardized units (lower is better). No autoregressive decoding and no teacher forcing is used. Causality is enforced by construction and verified empirically: a future-invariance and exact-zero-future-gradient suite passes for every variant. **[TODO: note: windows overlap by 255/256 positions and are therefore not independent samples; this is a declared limitation, see Section 5.3.]**

Training is identical for every configuration: AdamW (lr$=10^{-4}$, weight decay 10^{-5}), gradient clipping at norm 10, batch size 32, a fixed budget of 10 epochs, float32 on CPU (Apple Silicon, 8 threads), and deterministic seeding with the same seed used for data loading and model initialization. The fixed 10-epoch budget is justified by learning-curve analysis: all cells are in a plateau regime after epoch 4 and the mean last-3-epoch drift across all 20 runs is $+0.0009$; no cell keeps improving systematically, so the budget does not bias cell comparisons.

3.2 Model family

All configurations share one minimal Transformer block (hidden size 128, 4 heads, head dimension 32) with a horizon head $\mathrm{Linear}(128 \to 96 \cdot 7)$ applied to the last hidden state:

$$x = \mathrm{input_proj}(u) + \mathrm{pos_embedding}, \qquad h = \mathrm{MLP}(\mathrm{LN}(\mathrm{attn}(x) + x)), \qquad \hat{y} = \mathrm{horizon_head}(h[:, -1, :]), \quad (1)$$

The only difference between configurations is the attention module. The shared Q/KV/O projections follow the MLA-style pattern: a low-rank query projection with RMSNorm, a tied single-head KV projection (K = V) with RoPE applied to the last 8 of 32 dimensions, and a grouped O projection. A full causal attention variant (*Normal*) with no compression serves as the task-appropriate baseline.

3.3 The compressed-and-indexed attention family

Each compressed configuration consists of three components, all inherited verbatim from the reference implementation:

Two-series compressor. The L positions are projected to double width and gated by a per-dimension softmax over $2m$ slots, so that every token contributes to two adjacent windows (its own and the next); the weighted sum is RMSNormed and RoPE-encoded at block starts. With rate m, this yields $C = L/m$ KV blocks covering $[mj, mj + m - 1]$. The same compressor, with its own parameters, produces the $M = L/m_{idx}$ routing index tokens.

Router. Each query scores all M index tokens ($\mathrm{ReLU}(q \cdot \mathrm{idx}^\top)/\sqrt{d}$, head-mixed by a learned weight), masks tokens that lie in the future, and selects the top-k ($k{=}16$). In all compressed configurations the router reuses the main query (shared-Q), valid because the query head dimension equals the index head dimension; this removes one projection and one RoPE application.

Dense masked attention over the sparse selection. Instead of gathering the selected blocks, attention is computed densely over a fixed column set—the C KV blocks plus a sliding window of $W{=}16$—with $-\infty$ masking of unselected blocks and a single learnable "sink" softmax column for stability. Masking to $-\infty$ makes the dense result exactly equal to the sparse top-k selection, with no approximation. On CPU this formulation is several times faster than per-query gather in our implementation, and it keeps the softmax exact.

3.4 The two design axes and the factorial

The two axes are the KV compression rate $c_{KV} = m$ and the routing resolution $c_R = m_{idx}$. We evaluate the 2×2 factorial over $c_{KV}, c_R \in \{4{:}1, 8{:}1\}$, i.e. cells $(4, 4)$, $(8, 4)$, $(4, 8)$, $(8, 8)$. Table 1 lists the resulting geometry and parameter counts.

Parameter equalization. The factorial is pre-registered with an auxiliary-identical contract: all cells share window, top-k, heads, hidden size, optimizer, and schedule, and differ only in (c_{KV}, c_R)–the cell definition. Because the position-bias terms of the two compressors grow with their rates, attention-parameter counts differ slightly between

Table 1: Configuration geometry and parameters (measured; attention module and full model). All compressed cells share the auxiliary contract (window 16, top-k 16, 4 heads, head dimension 32, hidden 128, shared-Q, q/o ranks 64) and differ only in (c_{KV}, c_R); the low-rank bottlenecks are equalized to within 0.27% of the $(8, 4)$ reference. *Softmax cols* $= C + W + 1$ sink. Normal attends to all 256 columns with no sink.

Config	c_{KV}	c_R	KV blocks C	Index tokens M	Softmax cols	Params (attn / full)
Normal	—	—	—	—	256	102,496 / 355,200
$(4, 4)$	4	4	64	64	80 + 1	95,396 / 348,100
$(8, 4)$	8	4	32	64	48 + 1	95,652 / 348,356
$(4, 8)$	4	8	64	32	80 + 1	95,652 / 348,356
$(8, 8)$	8	8	32	32	48 + 1	95,908 / 348,612

Table 2: Factorial results (5 paired seeds, frozen direct-96 protocol; pooled MSE in standardized units, lower is better). "Beats (4,4)" counts seeds where the cell's validation MSE is better than the same seed's $(4, 4)$ value.

Cell	Val mean $\pm$ std	Test mean	Train mean	Beats (4,4)
$(4, 4)$	0.6332 ± 0.0611	0.9079	0.2908	—
$(8, 4)$	$\mathbf{0.5967} \pm 0.0710$	**0.7988**	0.2880	3 / 5
$(4, 8)$	0.6858 ± 0.0875	0.8996	0.2983	1 / 5
$(8, 8)$	0.6368 ± 0.0393	0.9159	0.2844	3 / 5

cells; the only free compute-quality knobs not part of the cell definition (the low-rank q and o bottlenecks) are fixed at the champion ranks $(64, 64)$, which is the lattice minimum for *every* cell. The largest remaining attention-parameter delta is 0.268% (full-model delta $\leq 0.074\%$), well inside the pre-registered 0.5% tolerance.

Complexity. Attention cost per query is proportional to the number of softmax columns: 256 for Normal, 80 for c_{KV}=4 cells, and 48 for c_{KV}=8 cells (5.3× fewer than Normal). Routing cost is $O(L \cdot M)$; with c_R=4, M=64 index tokens, identical to the c_R=4 cells regardless of c_{KV}. These are structural complexity facts; measured wall-clock latency is deferred to follow-up hardware benchmarking **[TODO: Gate 5: latency (p50/p90/p99, bootstrap CIs), FLOPs and memory accounting, cross-hardware (Apple Silicon + x86 CPU, GPU)]**.

4 Experiments

4.1 Setup and data integrity

The factorial comprises 4 cells $\times$ 5 paired seeds $\{42, 123, 2024, 7, 99\} = 20$ runs under the frozen protocol of Section 3.1. Data-integrity checks on the audited results manifest confirm: (i) parameter equalization within tolerance (max attention delta 0.268%, max full-model delta 0.074%); (ii) matched training across all runs (final training MSE span 0.2821–0.3002); and (iii) all 20 runs present, all 5 seeds in every cell, no duplicates. Reported validation and test metrics come from the epoch-10 weights; no best-validation checkpoint selection is used.

4.2 RQ1: the 2×2 factorial

Table 2 and Figure 1 summarize validation MSE by cell. The champion cell is $(8, 4)$ (mean validation 0.5967 ± 0.0710; mean test 0.7988), matching the corner that combines aggressive KV compression with fine routing. The worst cell is $(4, 8)$ (0.6858 ± 0.0875), which combines mild KV compression with coarse routing.

Main effects. Averaging over the opposite axis (Figure 2): coarsening KV compression from 4:1 to 8:1 *lowers* validation MSE by -0.0427 (mildly beneficial), while coarsening routing resolution from 4:1 to 8:1 *raises* it by $+0.0463$ (deterioration). Both signs are consistent in 7 of 10 seed-column comparisons each.

Pre-registered hypothesis H1 asked whether coarsening the router deteriorates performance *more* than coarsening KV compression. The strict form is *not* confirmed: the two effects are nearly equal in magnitude (0.0463 vs. 0.0427) and opposite in sign, so all observed deterioration is attributable to routing while KV compression is mildly beneficial. Both effect sizes ($\sim$0.04) are smaller than the mean within-cell seed standard deviation (0.0647; per-cell range 0.0393–0.0875), and no formal significance test was run. We therefore report the decoupling direction as *directional evidence*, not a conclusive result **[TODO: Gate 4: mixed-effects model** $\mathrm{MSE} = \beta_0 + \beta_1 \log_2 c_{KV} + \beta_2 \log_2 c_R + \beta_3 \log_2 c_{KV} \log_2 c_R +$ **random effects; the main claim requires** $\beta_2 > \beta_1$**; TOST/ROPE for equivalence; paired**

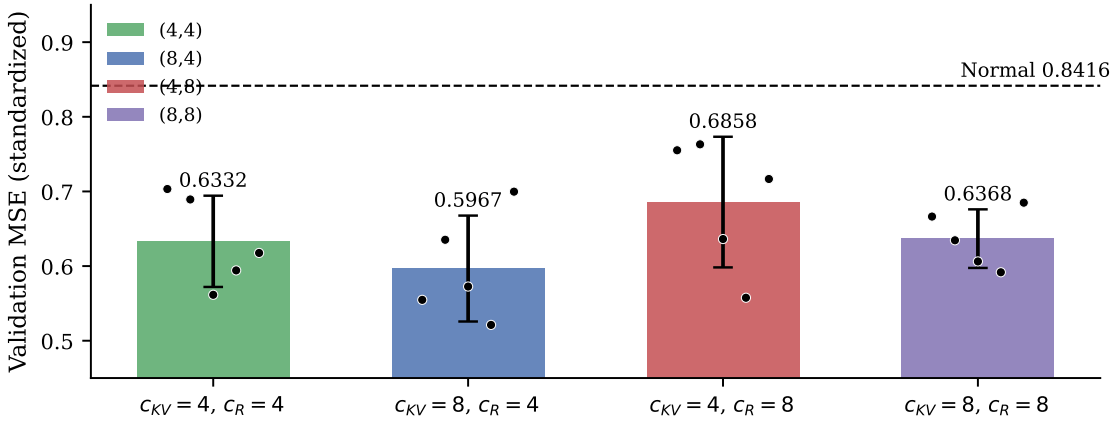


Figure 1: Validation MSE by factorial cell (mean ± std over 5 paired seeds; individual seed points shown). Lower is better. The champion corner (c_{KV}=8, c_R=4) combines coarse KV compression with fine routing.

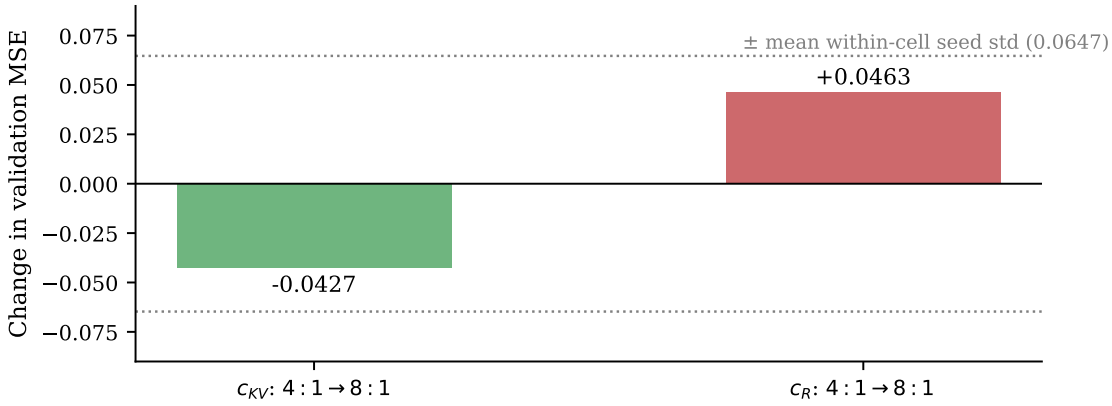


Figure 2: Main effects on validation MSE: c_{KV} 4:1 → 8:1 (-0.0427, mildly beneficial) and c_R 4:1 → 8:1 ($+0.0463$, deteriorates). Dashed lines: ± mean within-cell seed std (0.0647); both effects are smaller than seed noise.

Wilcoxon/Friedman with Holm correction]. Per-seed trajectories (Figure 3) show the ordering across cells is not unanimous for every seed—another reason the champion claim is stated as "best on mean," not "best for every seed."

4.3 RQ2: compressed vs. full attention

Table 3 compares full causal attention (Normal, 256 columns, no compression) with the two cells that anchor the family's design corners, $(4, 4)$ and the champion $(8, 4)$; Figure 3 shows all four cells. Every compressed cell beats Normal on mean validation and mean test MSE, and *every one of the 20 paired seed comparisons* favors the compressed configuration (range of validation deltas -18.5% to -29.1%; the $(8, 4)$ cell's advantage is the largest in 3 of 5 seeds, with $(4, 4)$ ahead at seeds 2024 and 99).

Wiring verification. Because the compressed variants beating an unmodified baseline by $\sim$25–29% is a surprising direction, we audited the Normal attention implementation before reporting it: a from-scratch naive reference (true causal masking, $1/\sqrt{d}$ scaling, independent RoPE reimplementation) matches the module output bit-exactly at L=32 and L=64 (0.0 max absolute difference) and to 1.19×10^{-7} at L=256 (fp32 accumulation noise); future positions receive exactly zero weight and future-token perturbation leaves outputs unchanged. The baseline is correctly implemented, so the comparison stands. We also note the direction is surprising in the other sense: compression acting as a task-appropriate inductive bias (see Section 5) is a plausible but *unverified* explanation **[TODO: H4 (pre-registered): mechanistic analysis of what the router attends to; change-point/event-based analyses]**.

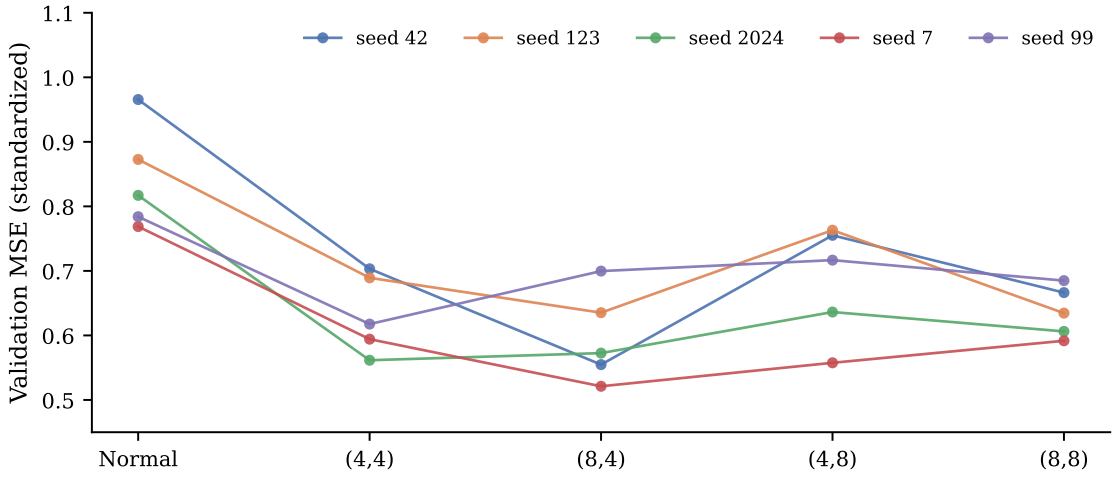


Figure 3: Per-seed validation MSE across configurations (parallel coordinates, one line per seed). All compressed cells beat Normal for every seed; the relative ordering of compressed cells varies by seed.

Table 3: Compressed vs. full attention (5 paired seeds, frozen protocol). Deltas are computed on the 4-decimal means. Parameter counts are not equalized across the three rows: Normal is a different attention class and carries ∼2% more full-model parameters.

Config	Val mean ± std	Test mean	Δval vs. Normal	Δtest vs. Normal	Seeds beating Normal
Normal	0.8416 ± 0.0799	1.0814	—	—	—
$(4, 4)$	0.6332 ± 0.0611	0.9079	-24.76%	-16.04%	5 / 5
$(8, 4)$	$\mathbf{0.5967} \pm 0.0710$	**0.7988**	-29.10%	-26.13%	5 / 5

4.4 Efficiency

Table 1 gives the structural efficiency story: the champion attends to 48 columns per query (5.3× fewer than Normal) at 95,652 attention parameters—*fewer* than Normal's 102,496—and a full-model count 2.0% below Normal's. Routing cost is identical for the two c_R=4 cells regardless of c_{KV}, which is the deliberate trade-off: pay the full fine-routing cost, halve the attention cost. Measured wall-clock latency, throughput, FLOPs, and memory are not yet reported for the frozen protocol; the earlier pilot-era speed figures are superseded and intentionally absent here **[TODO: Gate 5 hardware/latency benchmarking; see plan/next_step.md §10]**.

5 Discussion

5.1 The decoupling insight (directional)

The cleanest reading of the factorial is that the two axes play different roles: *routing granularity carries accuracy; KV content density only buys efficiency*. The control logic is internal to the design: the $(8, 4)$ cell (coarse content, fine router) matches or beats the equal-rate cells with half the attention columns, while the $(4, 8)$ cell (fine content, coarse router) is the worst cell despite having the same columns as $(4, 4)$. The per-seed consistency (7/10 in the same direction for each effect) and the effect-size-to-seed-noise comparison (both effects ∼0.04 vs. within-cell std ∼0.065) mean we currently report this as a *directional* finding; significance testing and a multi-dataset confirmation are the immediate next steps before this can be stated as a general design principle.

5.2 Why might compression help on this task?

We do not have a verified explanation for the compressed > full-attention outcome, and we explicitly do not claim one. Candidate (unverified) explanations include: compression acting as an inductive-bias regularizer; the block structure effectively smoothing the attention field over multi-step correlations; and the fixed 10-epoch budget interacting with optimization landscape differences. Each is testable, and the pre-registered hypothesis program in the project

plan is designed to separate them **[TODO: H1–H5 status per plan/paper_notes.md §5; mechanistic analyses are pending]**.

5.3 Limitations

- **Single dataset.** All results are on ETTh1 (7 features, the smallest of the ETT family) at CPU-scale model sizes. No cross-dataset or cross-scale generalization claim is made. **[TODO: Gate 3: ETTh2/ETTm1/ETTm2, Electricity, Traffic, Weather, Exchange; horizons $\{96, 192, 336, 720\}$]**.
- **Descriptive statistics.** Five paired seeds; no significance tests, confidence intervals beyond the reported std, or equivalence tests. "Best on mean" and "directional" are used advisedly.
- **No external baselines.** Comparisons are internal to the family plus the full-attention same-backbone baseline; PatchTST, iTransformer, DLinear, Informer, and linear/SSM baselines under the same protocol are pending **[TODO: Gate 3 baselines]**.
- **No latency or hardware results** for the frozen protocol; complexity claims are structural only **[TODO: Gate 5]**.
- **Non-IID samples.** Stride-1 windows overlap heavily; samples are not independent, which standard significance tests would assume **[TODO: Gate 4: block bootstrap / HAC-robust tests]**.
- **Fixed budget, single checkpoint policy.** 10 epochs, epoch-10 weights; no early stopping or best-validation selection.
- **CPU-only, fp32, small models.** Dense-with-masking implementation choices are CPU-oriented and may not transfer to GPU-flash-attention regimes.

5.4 What this paper does not claim

Consistent with the project's pre-registered warnings: this is not a novel sparse-attention mechanism; it is not a state-of-the-art forecasting system; the word "equivalence" is not used for any accuracy comparison; no subquadratic claim is made beyond the structural column counts above; and no component is claimed to be original. The contribution is the controlled study, its measurements, and its honest boundaries.

6 Conclusion

We presented a controlled 2×2 factorial design study of compressed-and-indexed attention for long-horizon time-series forecasting, varying the KV compression rate c_{KV} and the routing resolution c_R independently under a frozen task, an auxiliary-identical contract, and parameter equalization, across five paired seeds. On ETTh1 direct-96 forecasting, all compressed configurations outperformed full causal attention (-18.5% to -29.1% validation MSE; 20/20 paired seed comparisons; wiring-audited baseline), and the factorial separates the two axes directionally: coarser routing deteriorates accuracy while coarser KV compression is mildly beneficial, with the champion cell $(8, 4)$ cutting the attention matrix $5.3\times$ at equal parameters. We deliberately stop short of significance, generalization, or latency claims. The immediate roadmap is statistical testing with temporal-dependence-aware methods, multi-dataset replication, external baselines, and hardware latency benchmarking; each is itemized in the appendix roadmap (Appendix D).

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A Documented negative results

Six failed designs were tried and documented during the development of this family (each with runnable or archived implementations and failure modes): sparse-gather routing ($\sim$5$\times$ slower than dense masking on CPU); a hierarchical (multi-scale pyramid) indexer (consistently worse than the flat indexer); a multi-scale KV pyramid at 224 KV tokens (overparameterized for the budget); a learnable block-bias offset (NaN instability in $\sim$50% of runs; root cause later traced to uninitialized memory—'torch.empty' biases—and fixed); a linear-pool indexer with gated residual KV (loses the per-head $\times$ per-dimension routing flexibility of the softmax-overlap gate); and stacked 2:1 $\rightarrow$ 2:1 compression (harder to optimize within the budget). Negative results of this kind are a deliberate part of the reproducibility package; they are secondary to the paper's core claim.

B Surrogate attention (separate line of work)

A separate pipeline distills the attention block's output into a cheap student that replaces attention at inference: a per-token MLP student ($5.3\times$ faster, $+31\%$ validation MSE—per-token functions cannot replicate cross-token routing) and a windowed-plus-global-prototypes student ($2.18\times$ faster, $\sim 0.4\%$ gap to its teacher at a 5+1-epoch distillation schedule, $\sim 82\%$ fewer parameters than the teacher). These numbers predate the frozen task and are not apples-to-apples with the results in Section 4; the surrogate line is kept in this paper only to acknowledge the overlap with attention distillation work (Goldstein et al., 2025) and is intended for a separate publication **[TODO: decide retention: align distillation epochs to the main protocol and add an equal-budget small-transformer control (PUBLICATION.md §3.5 item 9)]**.

C Reproducibility

Every number in this paper traces to a repository artifact: the frozen task specification (the sole authority on task definition), the pre-registered factorial design, the run harness and its machine-readable results manifest (20 factorial runs + 5 Normal-baseline runs, JSONL), a verified per-module parameter registry, causality-verification and wiring-audit suites (all variants pass future-invariance and exact-zero-future-gradient checks; the Normal baseline is bit-identical to a naive causal reference), and a sanity-test suite covering shapes, gradients, parameter counts, and tiny-task fitting. Reported means and deltas are computed from the audited manifests; the figure-generation scripts in this paper's source tree read the same manifests, so every figure is regenerable. **[TODO: freeze-clean package before submission: environment lockfile, data hashes, automated table regeneration, unit tests as CI (plan/next_step.md §16)]**.

D Roadmap

1. **Statistics (Gate 4):** 5–10 paired seeds; mixed-effects model with $\beta_2 > \beta_1$ as the main claim; TOST or Bayesian ROPE for any equivalence statement; paired Wilcoxon/Friedman with Holm correction; block bootstrap or HAC-robust tests for temporal dependence.

2. **Datasets (Gate 3):** ETT family (ETTh2, ETTm1, ETTm2), large multivariate (Electricity, Traffic, Weather), different dynamics (Exchange, Solar-Energy, PEMS), plus a mechanistic synthetic; H and L sweeps.

3. **Baselines:** PatchTST, iTransformer, DLinear, Informer, TimesNet, plus a linear and an SSM baseline under the identical frozen protocol.

4. **Hardware/latency (Gate 5):** p50/p90/p99 with bootstrap CIs, throughput, FLOPs and memory accounting, scaling sweeps ($L \in \{128, \ldots, 4096\}$), and cross-hardware transfer (Apple Silicon, x86 CPU, GPU).

5. **Literature re-audit:** fresh 2024–2026 sweep of routing/compression-attention and TS-efficiency work before locking the title; resolve the outstanding citation-verification items flagged in the related-work section.